

EN.530.626: Optimal Control for Space Systems

Homework 1

Due Sep 19th 11:59PM

The code for this assignment can be found on GitHub at [this link](https://github.com/JHU-ACEL/ocss_hw) and can be downloaded by running `git clone https://github.com/JHU-ACEL/ocss_hw.git` from a terminal window. While you may directly download the repository as a zip file, we recommend that you clone it using `git` as we plan on releasing future problem sets and any bug fixes directly via this repository. Homework submissions and grading will be managed through Canvas.

Problem	Points Possible
Problem 1	15
Problem 2	15
Problem 3	15
Problem 4	15
Total	60

Introduction

This homework will focus on the following technical concepts:

1. Different schemes for functional differentiation (e.g., numerical, symbolic, automatic).
2. Solving simple optimization problems using gradient-based approaches.
3. Linear algebra concepts relating to quadratic expressions.

To accomplish this, the software development learning goals include,

- Writing Python code in an object-oriented fashion.
- Using a Docker environment for standardizing software packages.
- Using `marimo` notebooks for testing and prototyping code.

Setting up the Docker Environment

To set up the Docker image locally, follow the instructions located in `docker/README.md` instructions.

Once you have the `docker` container loaded, run,

```
$ python hw1/test_docker.py
```

and check if the packages are imported correctly.

Collaboration:

For each problem in this assignment, indicate in what capacity you collaborated with peers and which peers; if you did not collaborate, then include this information as well. **Your assignment will not be graded without this information.**

Problem 1: Matrix Math [15 Points]

Part A [6 Points]

Show that the ℓ_p -norm satisfies the properties of a metric norm for $p = 2$:

1. $\|\alpha x\|_2 = |\alpha| \|x\|_2$ for scalar α
2. $\|x\|_2 \geq 0$ for all x and $\|x\|_2 = 0$ if-and-only-if $x = 0$
3. $\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$

Hint: You may use the Cauchy-Schwartz inequality that for two vectors $a, b \in \mathbb{R}^n$, $(\langle a, b \rangle)^2 \leq (\langle a, a \rangle) \cdot (\langle b, b \rangle)$, where $\langle a, b \rangle$ is the inner product between the two vectors such that $\langle a, a \rangle = \|a\|_2^2$.

Part B [3 Points]

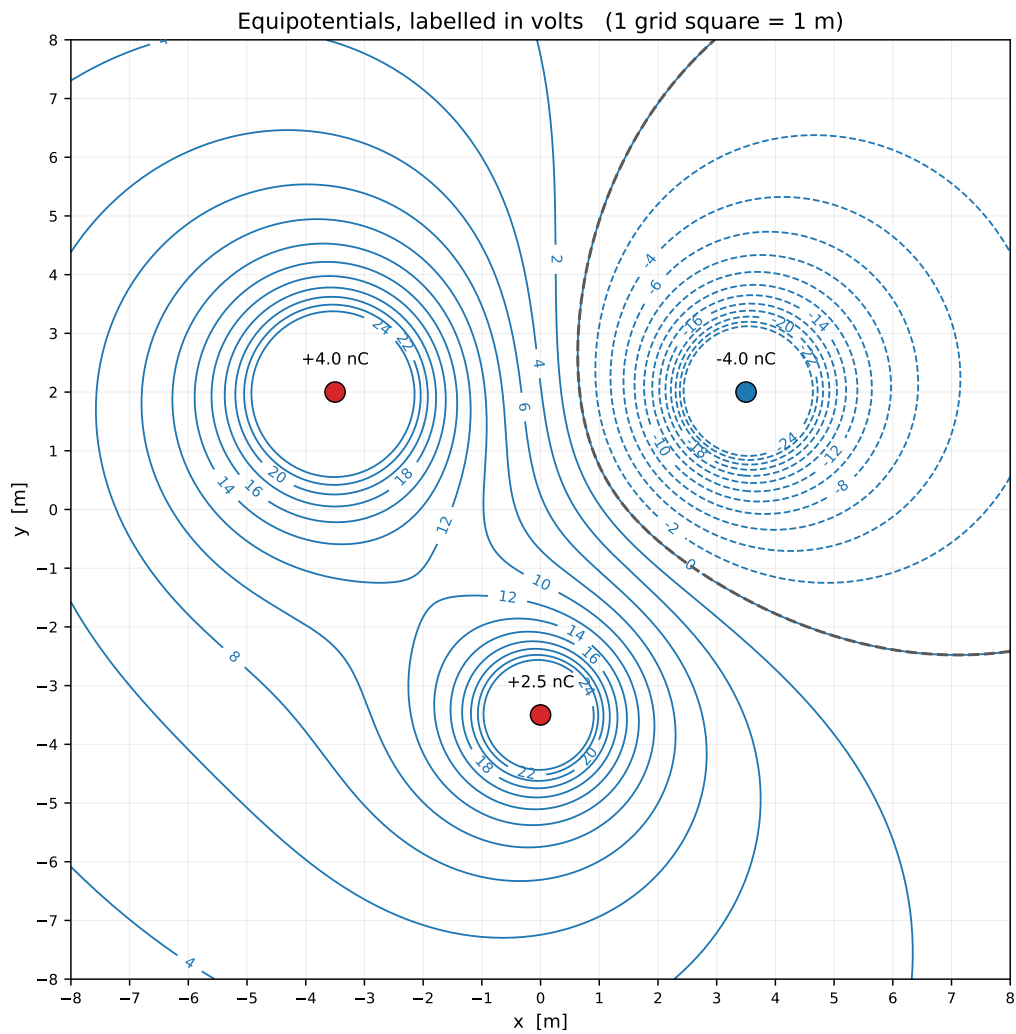
If p, q are positive integers (i.e., $p, q \in \mathbb{Z}_+$) such that $p > q \geq 1$, then we can show that $\|x\|_p \leq \|x\|_q$ for all $x \in \mathbb{R}^n$. Prove this for $p = 2$ and $q = 1$.

Part C [3 Points]

Suppose Q is a symmetric matrix (i.e., $Q \in \mathbb{S}^n$ such that $Q = Q^T$) and we claim $f(x) = x^T Q x$ attains a non-negative value for all non-zero x , i.e., $\forall x \in \mathbb{R}^n \setminus \{0\}, f(x) > 0$. Prove that all eigenvalues of Q are positive.

Part D [3 Points]

If an electric field $\vec{E}$ is conservative, then it can be written in terms of its electric potential V as $\vec{E} = -\nabla V$. Consider the following plot of equipotential curves generated by three charges. What is the electric field evaluated at $(0, 2)$?





Problem 2: Simple Optimization [15 Points]

In this problem, read the instructions provided in `hw1/problem_2.py`. There is no additional write-up required for this problem.

Part A [3 Points]

You will evaluate the gradient of a smooth and continuous function using analytical, numerical, and automatic differentiation techniques.

Part B [12 Points]

You will implement simple versions of gradient descent with line search and Newton method.

Problem 3: Least Squares

In this problem, read the instructions provided in `hw1/problem_3.py`. Further, there is additional write-up required to complete Part B.

Part A [5 Points]

You will use the analytical solution for linear least squares.

Part B [5 Points]

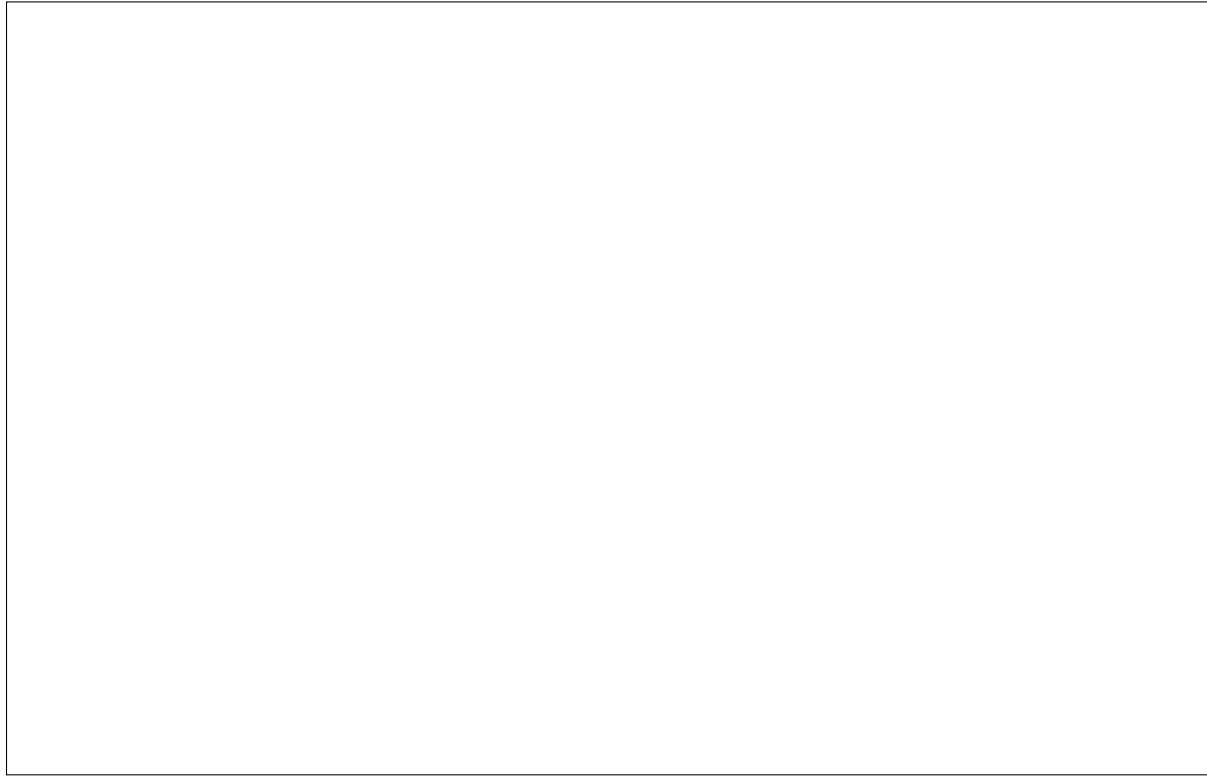
Consider the non-linear least squares (NLLS) problem of the form,

$$\min_x \frac{1}{2} \|f(x) - b\|_2^2,$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ (where $m > n$), $f \in \mathcal{C}^2$, and $b \in \mathbb{R}^m$.

Given an initial reference point $x^{(k)} \in \mathbb{R}^n$, derive what is known as the Gauss-Newton update rule to compute $x^{(k+1)}$ for the NLLS problem.

Hint: The update rule entails evaluating the linear Taylor series expansion of $f(x)$ (and not the quadratic expansion) by computing its Jacobian J about $x^{(k)}$.

**Part C [5 Points]**

You will implement the Gauss-Newton solver and use it for training a neural network.

Problem 4: An Iterative Linear Solver [15 Points]

In this problem, read the instructions provided in `hw1/problem_4.py`. There is no additional write-up required for this problem.

Part A [10 Points]

You will implement the preconditioned conjugate gradient (PCG) linear solver.

Part B [5 Points]

You will test the performance sensitivity of the PCG solver as a function of its condition number.

Submission Instruction

- Download the marimo notebook for each problem i.e 2, 3 and 4 and join them together into a single pdf named `hw1.pdf`.
- Compress the `hw1` folder containing your python files. Name this file “`hw1.zip`”.
- Upload both the PDF/HTML file and the zip folder into the canvas assignment “Homework 1”.

References